

Quantum Field Theory: Problem Set 2

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1 Emission, Absorption, and Thermalization in a Thermal Bath

Consider a probe particle of energy $\omega_{\mathbf{k}}$ with annihilation operator $a_{\mathbf{k}}$. The probe is weakly coupled to a thermal bath at temperature

$$T = \frac{1}{\beta}$$

through the interaction

$$H_{\text{int}} = g a_{\mathbf{k}}^{\dagger} O + g O^{\dagger} a_{\mathbf{k}},$$

where O is an operator acting on the Hilbert space of the thermal bath.

Throughout this problem, the bath is assumed to remain in thermal equilibrium and is characterized by the thermal correlation functions

$$G_{>}(t) = \langle O(t) O^{\dagger}(0) \rangle_{\beta}, \quad G_{<}(t) = \langle O^{\dagger}(0) O(t) \rangle_{\beta}.$$

Their Fourier transforms are defined by

$$G_{\gtrless}(\omega) = \int_{-\infty}^{\infty} dt e^{i\omega t} G_{\gtrless}(t).$$

1.1 Emission and absorption rates

Consider first a single probe mode. Denote by

$$|0\rangle$$

the probe vacuum and by

$$|1_{\mathbf{k}}\rangle = a_{\mathbf{k}}^{\dagger} |0\rangle$$

the one-particle state.

Emission. The interaction term

$$a_{\mathbf{k}}^\dagger O$$

creates a probe particle while extracting energy from the thermal bath.

Using first-order time-dependent perturbation theory (or Fermi's golden rule), compute the transition rate

$$|0\rangle \longrightarrow |1_{\mathbf{k}}\rangle.$$

Show that

$$\boxed{\Gamma_{0 \rightarrow 1} = g^2 G_{<}(\omega_{\mathbf{k}})}$$

up to the conventional normalization associated with the definition of the Fourier transform.

Explain why $G_{<}(\omega)$ may be interpreted as the emission spectrum of the thermal bath.

Absorption. The interaction term

$$O^\dagger a_{\mathbf{k}}$$

destroys a probe particle while depositing energy into the thermal bath.

Compute the transition rate

$$|1_{\mathbf{k}}\rangle \longrightarrow |0\rangle.$$

Show that

$$\boxed{\Gamma_{1 \rightarrow 0} = g^2 G_{>}(\omega_{\mathbf{k}})}$$

up to the same normalization convention.

Explain why $G_{>}(\omega)$ may be interpreted as the absorption spectrum of the thermal bath.

1.2 Evolution of an occupied mode

Now consider a probe mode containing $n_{\mathbf{k}}$ particles.

To determine the time evolution of the occupation number, consider separately the transitions

$$n_{\mathbf{k}} \rightarrow n_{\mathbf{k}} + 1$$

generated by $a_{\mathbf{k}}^\dagger O$, and

$$n_{\mathbf{k}} \rightarrow n_{\mathbf{k}} - 1$$

generated by $O^\dagger a_{\mathbf{k}}$.

Bosonic probe.

1. Show that

$$\frac{\langle n+1|a^\dagger|n\rangle}{\sqrt{\langle n+1|n+1\rangle}\sqrt{\langle n|n\rangle}} = \sqrt{n+1}, \quad \frac{\langle n-1|a^\dagger|n\rangle}{\sqrt{\langle n-1|n-1\rangle}\sqrt{\langle n|n\rangle}} = \sqrt{n}.$$

2. Hence show that the corresponding transition rates are proportional to

$$(n_{\mathbf{k}} + 1) G_{<}(\omega_{\mathbf{k}}),$$

and

$$n_{\mathbf{k}} G_{>}(\omega_{\mathbf{k}}),$$

respectively.

3. Deduce that

$$\boxed{\frac{dn_{\mathbf{k}}}{dt} = g^2 \left[(1 + n_{\mathbf{k}}) G_{<}(\omega_{\mathbf{k}}) - n_{\mathbf{k}} G_{>}(\omega_{\mathbf{k}}) \right]},$$

4. Explain the physical origin of the Bose enhancement factor $1 + n_{\mathbf{k}}$.

Fermionic probe.

1. Repeat the analysis for a fermionic probe satisfying

$$(a_{\mathbf{k}}^\dagger)^2 = 0.$$

2. Show that

$$\boxed{\frac{dn_{\mathbf{k}}}{dt} = g^2 \left[(1 - n_{\mathbf{k}}) G_{<}(\omega_{\mathbf{k}}) - n_{\mathbf{k}} G_{>}(\omega_{\mathbf{k}}) \right]}.$$

3. Explain the physical origin of the Pauli blocking factor $1 - n_{\mathbf{k}}$.

1.3 Detailed balance and thermal equilibrium

We have shown in class the thermal bath satisfies the KMS relation

$$G_{>}(\omega) = e^{\beta\omega} G_{<}(\omega).$$

Bosonic probe. Determine the stationary solution for which

$$\frac{dn_{\mathbf{k}}}{dt} = 0$$

and show that it is given by the Bose-Einstein distribution

$$n_{\mathbf{k}} = \frac{1}{e^{\beta\omega_{\mathbf{k}}} - 1}.$$

Fermionic probe. Show that in the stationary solution we should have the Fermi-Dirac distribution.

$$n_{\mathbf{k}} = \frac{1}{e^{\beta\omega_{\mathbf{k}}} + 1}.$$

1.4 Physical interpretation

1. Explain why the Bose-Einstein and Fermi-Dirac distributions are precisely the distributions for which emission and absorption satisfy detailed balance.
2. Discuss the roles of Bose enhancement and Pauli blocking in determining the equilibrium occupation numbers.

2 Analytic Structure of Thermal Correlation Functions

Consider a bosonic or fermionic operator $O(t)$ in thermal equilibrium at temperature

$$T = \frac{1}{\beta}.$$

Define the thermal Wightman functions

$$G_{>}(t) = \langle O(t)O^{\dagger}(0) \rangle_{\beta},$$

$$G_{<}(t) = \langle O^{\dagger}(0)O(t) \rangle_{\beta}.$$

Their Fourier transforms are

$$G_{\gtrless}(\omega) = \int_{-\infty}^{\infty} dt e^{i\omega t} G_{\gtrless}(t).$$

The retarded, advanced, and Feynman correlators are defined by

$$G_R(t) = -i\theta(t)(G_{>}(t) \mp G_{<}(t)),$$

$$G_A(t) = i\theta(-t)(G_>(t) \mp G_<(t)),$$

$$G_F(t) = -i\langle T O(t) O^\dagger(0) \rangle_\beta,$$

where the upper sign refers to bosons and the lower sign to fermions.

2.1 Spectral representations and KMS relation

1. Insert a complete set of energy eigenstates and derive the Lehmann representations

$$G_<(\omega) = \frac{2\pi}{Z} \sum_{m,n} e^{-\beta E_n} |O_{mn}|^2 \delta(\omega - E_n + E_m)$$

and

$$G_>(\omega) = \frac{2\pi}{Z} \sum_{m,n} e^{-\beta E_n} |O_{mn}|^2 \delta(\omega - E_m + E_n).$$

2. By interchanging the dummy indices m and n , show that

$$G_>(\omega) = e^{\beta\omega} G_<(\omega)$$

for bosonic operators and

$$G_>(\omega) = -e^{\beta\omega} G_<(\omega)$$

for fermionic operators.

3. Using the KMS condition, show that

$$G_>(t) = \pm G_<(t - i\beta).$$

2.2 Thermal analyticity strips

Define the analytic continuations

$$G_{\geq}(z) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-i\omega z} G_{\geq}(\omega).$$

1. Review the argument discussed in class showing that $G_<(z)$ is analytic in the strip

$$0 < \text{Im } z < \beta.$$

2. Repeat the argument for $G_>(z)$ and show that it is analytic in the strip

$$\boxed{-\beta < \text{Im } z < 0.}$$

3. Show that the KMS condition identifies the upper boundary of the $G_>$ strip with the lower boundary of the $G_<$ strip.

2.3 Retarded and advanced correlators

Define the spectral density as the Fourier transform of the (anti-)commutator for (fermions) bosons:

$$\rho(\omega) = G_>(\omega) \mp G_<(\omega).$$

1. Show that

$$\boxed{G_R(\omega) = \int_{-\infty}^{\infty} \frac{d\omega'}{2\pi} \frac{\rho(\omega')}{\omega - \omega' + i\epsilon}}$$

and

$$\boxed{G_A(\omega) = \int_{-\infty}^{\infty} \frac{d\omega'}{2\pi} \frac{\rho(\omega')}{\omega - \omega' - i\epsilon}.$$

2. Determine the domains of analyticity of $G_R(\omega)$ and $G_A(\omega)$.
3. Show that

$$\rho(\omega) = i(G_R(\omega) - G_A(\omega)).$$

2.4 Feynman correlator at finite temperature

1. Starting from the definition of G_F , derive its spectral representation.
2. Show that for bosons

$$\boxed{G_F(\omega) = G_R(\omega) + n_B(\omega)[G_R(\omega) - G_A(\omega)],}$$

or equivalently,

$$\boxed{G_F(\omega) = (1 + n_B(\omega)) G_R(\omega) - n_B(\omega) G_A(\omega),}$$

where

$$n_B(\omega) = \frac{1}{e^{\beta\omega} - 1}.$$

3. Show that for fermions

$$G_F(\omega) = G_R(\omega) - n_F(\omega) [G_R(\omega) - G_A(\omega)],$$

or equivalently,

$$G_F(\omega) = (1 - n_F(\omega)) G_R(\omega) + n_F(\omega) G_A(\omega),$$

where

$$n_F(\omega) = \frac{1}{e^{\beta\omega} + 1}.$$

4. Verify that these expressions reduce to the zero-temperature relation

$$G_F(\omega) = \theta(\omega) G_R(\omega) + \theta(-\omega) G_A(\omega).$$

2.5 Zero-temperature analyticity and Wick rotation

In this subsection set $T = 0$. From the previous part, the Feynman correlator satisfies on the real frequency axis

$$G_F(\omega) = \theta(\omega) G_R(\omega) + \theta(-\omega) G_A(\omega).$$

1. Recall that $G_R(z)$ is analytic for $\text{Im } z > 0$, whereas $G_A(z)$ is analytic for $\text{Im } z < 0$. Explain why the positive-frequency boundary value of G_F therefore admits analytic continuation through the first quadrant of the complex frequency plane, while the negative-frequency boundary value admits analytic continuation through the third quadrant.
2. Consider a real-frequency integral involving the Feynman correlator. Show that the positive-frequency part of the contour can be rotated through the first quadrant to the positive imaginary axis, while the negative-frequency part can be rotated through the third quadrant to the negative imaginary axis, without crossing singularities.
3. Hence explain why one may perform the Wick rotation

$$\omega = i\omega_E$$

and thereby obtain the Euclidean two-point function from the zero-temperature Feynman correlator. State clearly which analyticity properties make this continuation possible.

4. Sketch the singularity structure in the complex ω -plane and the corresponding contour deformation. Explain why the argument is non-perturbative and does not rely on a Feynman-diagram expansion.

3 Causal Diffusion, Gradient Expansion, and Entropy Production

Consider a non-relativistic conserved charge density $n(t, \mathbf{x})$ and current $\mathbf{J}(t, \mathbf{x})$ satisfying

$$\partial_t n + \nabla \cdot \mathbf{J} = 0.$$

Assume that the current obeys the Israel–Stewart–Müller (Maxwell–Cattaneo) equation

$$\tau \partial_t \mathbf{J} + \mathbf{J} = -D \nabla n,$$

where D is the diffusion constant and τ is a relaxation time.

The goal of this problem is to understand how ordinary hydrodynamics emerges from this causal theory as a systematic derivative expansion, and how the resulting transport coefficients admit a thermodynamic interpretation.

3.1 Hydrodynamic derivative expansion

In the hydrodynamic regime, spatial gradients are small and diffusion implies the scaling

$$\omega \sim Dk^2.$$

Treat

$$\nabla = O(\epsilon), \quad \partial_t = O(\epsilon^2)$$

as formal derivative-counting rules.

1. Show that the current admits an expansion

$$\mathbf{J} = \mathbf{J}^{(1)} + \mathbf{J}^{(3)} + \mathbf{J}^{(5)} + \dots,$$

where the superscript denotes the number of derivatives according to the above counting.

2. Using the Israel–Stewart equation, determine the leading-order constitutive relation $\mathbf{J}^{(1)}$.
3. Keeping terms through $O(\epsilon^3)$, iteratively solve the Israel–Stewart equation for $\mathbf{J}$. Show that

$$\mathbf{J} = -D \nabla n + \tau D \partial_t \nabla n + O(\epsilon^5).$$

4. Use the leading-order conservation law together with the leading constitutive relation to derive the diffusion equation for n . Eliminate the time derivative in the previous result and rewrite the constitutive relation entirely in terms of spatial gradients of n .
5. Show that the constitutive relation takes the form

$$\mathbf{J} = -D\nabla n + \lambda \nabla \nabla^2 n + O(\epsilon^5),$$

and determine the higher-order transport coefficient λ in terms of D and τ .

6. By iterating the procedure once more, obtain the next correction proportional to $\nabla \nabla^4 n$. Determine the corresponding transport coefficient.
7. Show that the constitutive relation may be written formally as

$$\mathbf{J} = -D(1 + \tau \partial_t)^{-1} \nabla n,$$

and explain how the resulting infinite gradient expansion may be interpreted as arising from integrating out a non-hydrodynamic mode.

3.2 Entropy current and entropy production

Assume local thermodynamic equilibrium and constant temperature. The local entropy density $s(n)$ satisfies

$$ds = \frac{\mu}{T} dn,$$

where μ is the chemical potential.

1. Using charge conservation, show that

$$\partial_t s = -\frac{\mu}{T} \nabla \cdot \mathbf{J}.$$

2. Rearrange this expression into the form

$$\partial_t s + \nabla \cdot \mathbf{s} = \Sigma,$$

and determine an entropy current $\mathbf{s}$.

3. Show that the local entropy production rate is

$$\Sigma = \mathbf{J} \cdot \nabla \left(\frac{\mu}{T} \right).$$

- Using the leading-order constitutive relation, show that

$$\Sigma \geq 0$$

provided the diffusion constant is positive.

- Explain why only the leading-order constitutive relation is required in order to establish positivity of entropy production to the order considered.

3.3 Thermodynamic interpretation of the relaxation time

The hydrodynamic constitutive relation obtained in part (a) contains the higher-order transport coefficient

$$\lambda = \tau D^2.$$

One would like to understand the thermodynamic origin of the parameter τ .

Assume that away from strict local equilibrium the entropy density receives a quadratic correction,

$$s = s_{\text{eq}}(n) - \frac{\beta}{2} \mathbf{J}^2,$$

where β is a phenomenological coefficient.

Let

$$\chi = \left(\frac{\partial n}{\partial \mu} \right)_T$$

denote the static susceptibility.

- Recompute the entropy production rate using the modified entropy density.
- Show that positivity of entropy production is achieved if the current satisfies a constitutive relation of Israel–Stewart form.
- Determine the relation between β , D , τ , and χ .
- Explain the physical meaning of the coefficient β .
- Discuss how the relaxation time τ may be interpreted as the hydrodynamic remnant of a non-hydrodynamic mode that has been integrated out.

4 Retarded Correlation Functions in Causal Charge Diffusion

Consider a conserved charge density $n(t, \mathbf{x})$ and current $\mathbf{J}(t, \mathbf{x})$ satisfying

$$\partial_t n + \nabla \cdot \mathbf{J} = 0,$$

together with the Israel–Stewart–Müller constitutive relation

$$\tau \partial_t \mathbf{J} + \mathbf{J} = -D \chi \nabla \mu,$$

where

$$\chi = \left(\frac{\partial n}{\partial \mu} \right)_T$$

is the static susceptibility.

Throughout this problem, work to linear order in fluctuations around a homogeneous equilibrium state.

4.1 Density-density correlator

Couple the system to an external chemical potential source,

$$\delta H = - \int d^3x n \mu_{\text{ext}}.$$

Assume that

$$\delta n = \chi (\delta \mu + \mu_{\text{ext}}).$$

1. Fourier transform the conservation law and constitutive relation to $(\omega, \mathbf{k})$ -space and solve for $\delta n(\omega, \mathbf{k})$ in terms of $\mu_{\text{ext}}(\omega, \mathbf{k})$.
2. Using

$$\delta n = G_{nn}^R(\omega, k) \mu_{\text{ext}},$$

derive the retarded density-density correlator and show that

$$G_{nn}^R(\omega, k) = -\chi \frac{Dk^2}{Dk^2 - i\omega - \tau\omega^2}.$$

3. Determine the poles of G_{nn}^R and identify the hydrodynamic and non-hydrodynamic modes.

4.2 Current correlators from a vector source

Now couple the system to an external vector potential,

$$\delta H = - \int d^3x J_i A_i.$$

Assume that the constitutive relation becomes

$$\tau \partial_t J_i + J_i = -D\chi \partial_i \mu + \chi D E_i,$$

where

$$E_i = -\partial_t A_i.$$

Restrict attention to the longitudinal sector.

1. Solve the linearized hydrodynamic equations for n and J_L in the presence of the source A_L .
2. Using

$$\delta n = G_{nJ_L}^R(\omega, k) A_L,$$

derive the mixed correlator $G_{nJ_L}^R$.

3. Using

$$\delta J_L = G_{J_L J_L}^R(\omega, k) A_L,$$

derive the longitudinal current-current correlator.

4. Show that

$$G_{nJ_L}^R = G_{J_L n}^R = -\chi \frac{D\omega k}{Dk^2 - i\omega - \tau\omega^2},$$

and

$$G_{J_L J_L}^R = -\chi \frac{D\omega^2}{Dk^2 - i\omega - \tau\omega^2}.$$

4.3 Ward identities

Charge conservation implies the exact Ward identities

$$\omega G_{nn}^R = k G_{J_L n}^R,$$

$$\omega G_{n J_L}^R = k G_{J_L J_L}^R,$$

and

$$\omega^2 G_{nn}^R = k^2 G_{J_L J_L}^R.$$

Verify these identities explicitly using the correlators obtained above.

4.4 Static and dynamic limits

1. Evaluate

$$-\lim_{k \rightarrow 0} \lim_{\omega \rightarrow 0} G_{nn}^R(\omega, k),$$

and show that it reproduces the thermodynamic susceptibility χ .

2. Evaluate

$$-\lim_{\omega \rightarrow 0} \lim_{k \rightarrow 0} G_{nn}^R(\omega, k).$$

3. Explain physically why the two limits do not commute.

4.5 Causality and transport bounds

The poles of the retarded correlators satisfy

$$\tau\omega^2 + i\omega - Dk^2 = 0.$$

1. Determine the large- k asymptotic form of the dispersion relation.
2. Show that disturbances propagate with front velocity

$$v_{\text{front}} = \sqrt{\frac{D}{\tau}}.$$

3. Assuming an underlying relativistic microscopic theory, impose

$$v_{\text{front}} \leq c$$

and derive the bound

$$D \leq c^2 \tau.$$

4. Rewrite the result in natural units $c = 1$.
5. Define the dimensionless diffusivity

$$\hat{D} = DT.$$

Does causality alone imply a universal bound on $\hat{D}$?

6. Assuming additionally that

$$\tau \sim \frac{\hbar}{k_B T},$$

show that causality implies a bound on DT , and briefly discuss its physical significance.

5 Ward Identities, Gauge Invariance, and the Einstein Relation

Consider a non-relativistic many-body system with conserved charge density $n(t, \mathbf{x})$ and current density $J_i(t, \mathbf{x})$. The operators satisfy

$$\partial_t n + \partial_i J_i = 0.$$

The system is coupled to an external vector potential through

$$H[A] = H_0 - \int d^d x J_i A_i + \frac{1}{2m} \int d^d x n A^2.$$

Retarded correlators are defined by

$$G_{AB}^R(t, \mathbf{x}) = -i\theta(t) \langle [A(t, \mathbf{x}), B(0)] \rangle.$$

Define the paramagnetic current-current correlator

$$\Pi_{ij}(\omega, \mathbf{k}) = G_{J_i J_j}^R(\omega, \mathbf{k}),$$

and the full gauge-invariant response kernel

$$K_{ij}(\omega, \mathbf{k}) = \Pi_{ij}(\omega, \mathbf{k}) + \frac{\bar{n}}{m} \delta_{ij},$$

where

$$\bar{n} = \langle n \rangle$$

is the equilibrium charge density.

5.1 Paramagnetic and diamagnetic response

1. Using linear response theory, show that the induced current may be written as

$$\delta\langle J_i \rangle = -\Pi_{ij} A_j - \frac{\bar{n}}{m} A_i.$$

2. Identify the paramagnetic and diamagnetic contributions.
3. Explain briefly why the diamagnetic term is required by gauge invariance.

5.2 Ward identities for the paramagnetic correlators

Following the derivation discussed in class, use the continuity equation

$$\partial_t n + \partial_i J_i = 0$$

to show that

$$\omega G_{nn}^R = k_i G_{J_i n}^R,$$

$$\omega G_{n J_j}^R = k_i \Pi_{ij} + \frac{\bar{n}}{m} k_j.$$

and

$$\omega^2 G_{nn}^R = k_i k_j \Pi_{ij} + \frac{\bar{n}}{m} k^2.$$

Include all steps skipped in the class including the derivation of the Schwinger contact term in charge-current commutator.

5.3 Gauge-invariant Ward identities

1. Show that inclusion of the diamagnetic contribution removes the contact term appearing above.
2. Prove that the full response kernel satisfies

$$\boxed{\omega G_{n J_j}^R = k_i K_{ij},}$$

and

$$\boxed{\omega^2 G_{nn}^R = k_i k_j K_{ij}.}$$

3. Explain briefly why these are the gauge-invariant Ward identities.

5.4 Longitudinal correlators

Define

$$J_L = \frac{k_i J_i}{k}, \quad \Pi_{LL} = \frac{k_i k_j}{k^2} \Pi_{ij}, \quad K_{LL} = \frac{k_i k_j}{k^2} K_{ij}.$$

1. Show that the paramagnetic Ward identities reduce to

$$\omega G_{nn}^R = k G_{J_L n}^R,$$

$$\omega G_{n J_L}^R = k \Pi_{LL} + \frac{\bar{n}}{m} k,$$

and

$$\omega^2 G_{nn}^R = k^2 \Pi_{LL} + \frac{\bar{n}}{m} k^2.$$

2. Show that the full Ward identities reduce to

$$\omega G_{n J_L}^R = k K_{LL},$$

and

$$\omega^2 G_{nn}^R = k^2 K_{LL}.$$

5.5 Hydrodynamic solution

From hydrodynamics you have derived that the long-wavelength density-density correlator takes the diffusive form

$$G_{nn}^R(\omega, \mathbf{k}) = -\chi \frac{Dk^2}{Dk^2 - i\omega}.$$

1. State your earlier results for how Ward identities determine

$$G_{J_L n}^R, \quad G_{n J_L}^R, \quad K_{LL}.$$

2. Using the paramagnetic Ward identity, determine

$$\Pi_{LL}.$$

3. Show that

$$\Pi_{LL}(\omega, k) = -\frac{\bar{n}}{m} - \chi \frac{D\omega^2}{Dk^2 - i\omega},$$

and verify explicitly that

$$K_{LL} = \Pi_{LL} + \frac{\bar{n}}{m}.$$

Here you can be brief as you have already done most of these in the earlier problems.

5.6 Conductivity and the Einstein relation

Consider a spatially uniform electric field

$$\mathbf{E}(\omega) = i\omega \mathbf{A}(\omega), \quad \mathbf{k} = 0.$$

The conductivity is defined through

$$J_i(\omega) = \sigma(\omega) E_i(\omega).$$

1. Show that linear response implies

$$\sigma(\omega) = -\frac{1}{i\omega} K_{LL}(\omega, \mathbf{k} = 0).$$

2. Show that

$$\Pi_{LL}(\omega, 0) = -\frac{\bar{n}}{m} - i\chi D\omega + O(\omega^2).$$

3. Explain why the constant term is completely fixed by the paramagnetic Ward identity.

4. Show that this constant term is exactly cancelled in the gauge-invariant response kernel and obtain

$$K_{LL}(\omega, 0) = -i\chi D\omega + O(\omega^2).$$

5. Hence derive Einstein's relation

$$\boxed{\sigma = \chi D.}$$

6. Explain why omitting the diamagnetic contribution would lead to an unphysical result for the conductivity.

5.7 Drude conductivity from causal diffusion

In the Israel–Stewart theory discussed previously, the current obeys

$$\tau \partial_t J_i + J_i = \sigma E_i.$$

1. Consider a spatially uniform electric field ($k = 0$) and solve for the current response in frequency space.
2. Show that the frequency-dependent conductivity is

$$\sigma(\omega) = \frac{\sigma}{1 - i\omega\tau}.$$

3. Determine

$$\text{Re } \sigma(\omega), \quad \text{Im } \sigma(\omega).$$

4. Verify that

$$\lim_{\omega \rightarrow 0} \sigma(\omega) = \sigma.$$

5. Using Einstein’s relation, rewrite the Drude conductivity entirely in terms of D , χ , and τ .

5.8 Physical interpretation of Einstein’s relation

The hydrodynamic current is

$$\mathbf{J} = -D\nabla n + \sigma \mathbf{E}.$$

Using

$$\delta n = \chi \delta \mu, \quad \mathbf{E} = -\nabla \phi,$$

define the electrochemical potential

$$\mu_{\text{tot}} = \mu + \phi.$$

1. Rewrite the current in terms of μ and μ_{tot} .
2. Show that if

$$\mu_{\text{tot}}(\mathbf{x}) = \text{constant},$$

the current vanishes if and only if

$$\sigma = \chi D.$$

3. Interpret the two terms in the current as drift and diffusion currents.
4. Explain physically why Einstein's relation guarantees exact cancellation of drift and diffusion in thermal equilibrium.